TRACE-X System Architecture

Explainable GNN risk prediction on the Elliptic Bitcoin dataset · proposed design

Module · Data store · Request / control flow · Data / model artifact · Audit / lineage events · Core contribution (L3) · Workflow step (1–10)

OFFLINE · DATA & TRAINING

L1 · DATA PREPARATION

Elliptic Dataset

CSV

1 Dataset Loader
features · labels
timestamps · edges

raw

2 Validation
clean · scale · balance
time-aware splits

clean

3 Graph Builder
tx = node
flow = edge

PyG

Graph Store

graph + features

L2 · GRAPH LEARNING & PREDICTION

Baselines
LogReg · RandomForest
same splits

4 GraphSAGE
neighbour aggregation
PyTorch Geometric

Evaluation
P · R · F1 · PR-AUC
threshold

best

Model Registry

model weights + config

ONLINE · SERVICE

L4 · BACKEND API · FastAPI · PyTorch · PyG · hosts L2 inference and L3 explainability

API Router
auth · validation
error handling

tx id

5 Prediction Service
GraphSAGE inference
risk score + threshold

scores

L3 · EXPLAINABILITY & INVESTIGATION (CORE)

6 Case Selection
rank by risk
review policy

picked

7 GNNExplainer
subgraph + features
per prediction

expl.

8 Explanation Validator
faithfulness · stability
flags weak output

checked

9 Evidence Assembly
id · class · score
subgraph · features

model + graph context

REST / JSON

ranked cases

evidence record · JSON

CLIENT

L5 · INVESTIGATOR INTERFACE · React · TypeScript · Cytoscape.js

Investigator

Investigation Queue
ranked cases
search · filter

select

Graph Explorer
selected tx
neighbourhood

inspect

Explanation Panel
subgraph · features
quality warnings

assess

10 Case Review
confirm · dismiss
escalate · notes

export

Case Record
JSON export
not a legal verdict

PLATFORM

L6 · CROSS-CUTTING SECURITY & RELIABILITY

events · lineage · metrics

Access Control
authN/Z · scoped actions

Data Lineage
datasets · steps · models

Reproducibility
configs · params · results

Audit Log
predictions · reviews

Model Monitoring
accuracy · explanation quality

Human Oversight
investigator decides

Proposed architecture. Until the components are implemented and tested, model performance, explanation quality and investigation outcomes are planned evaluations, not measured results.

A risk score is a model prediction, not proof of fraud · An explanation shows influencing patterns, not causation · Investigator actions are assessments, not legal conclusions.